

Driver Registration, Compliance & Logistics API Specification

Target Audience: Mobile Developers (React Native/Flutter), Frontend Engineers, Backend Core, Logistics Compliance

Base URL: <https://api.bemsfarms.com> | **Auth:** JWT Bearer Token | **Version:** 2.0 (Production)

1. System Overview & Onboarding Lifecycle

Bems Farms implements an end-to-end **Driver Invitation, Self-Service Document Submission, and Compliance Review Workflow**. Every driver on the platform undergoes verification of their National Identity Number (NIN), Driver's License, Vehicle Registration Papers, and Guarantor details before activation.

Phase	Actor	Key Action & System Result
Phase 1: Invitation	Admin Dispatch	Admin invites driver via email. System generates a secure 7-day single-use token and temporary PIN, sending an invitation email.
Phase 2: Submission	Driver Candidate	Driver opens onboarding portal, enters Next of Kin, Guarantor, and Payout details, and uploads document scans (License, NIN, Vehicle Papers, Photo).
Phase 3: Review	Admin Compliance	Admin inspects candidate dossier & document scans in the Compliance Modal. Admin either approves or requests corrections with notes.
Phase 4: Activation	Driver Mobile App	Driver receives approval email with Phone & PIN. Driver signs into the Mobile App, toggles shift Online, and starts receiving order dispatches.

2. Onboarding Status States (`onboarding\_status`)

Status Enum	Driver Status	Operational Meaning
invited	pending	Invite email sent. Awaiting driver to access onboarding portal.
documents_submitted	pending	Driver completed forms & uploaded scans. Pending admin compliance check.
approved	active	Compliance approved. Driver credentials active for mobile app login.
rejected	suspended	Compliance rejected. Rejection email sent to driver with corrective notes.

3. Database Schema Reference (`drivers` Table)

Column	Type	Description
id	SERIAL PK	Unique driver primary key.
name, phone, email	VARCHAR	Driver full name, unique phone (login ID), and email address.
nin_number	VARCHAR(50)	11-digit National Identification Number.
vehicle_type, plate	VARCHAR(50)	Vehicle type (motorcycle/van/car) and registration plate.
emergency_contact_*	VARCHAR	Next of Kin Name, Phone, and Relationship.
guarantor_*	VARCHAR / TEXT	Guarantor Full Name, Phone, and Residential/Work Address.
documents	JSONB	Object with URLs: { drivers_license, nin_slip, vehicle_registration, passport_photo, guarantor_form }.
onboarding_status	VARCHAR(50)	invited documents_submitted approved rejected.
compliance_notes	TEXT	Admin feedback notes / required correction details.
invite_token, expires_at	VARCHAR / TS	Cryptographic invite token and 7-day expiration timestamp.
commission_per_delivery	NUMERIC(10,2)	Custom driver commission per completed trip (default ■500).

4. Admin Dispatch & Compliance API Endpoints

4.1 POST /api/admin/deliveries/drivers/invite

Description: Sends onboarding invitation email to candidate with 7-day token link.

Auth: Admin JWT Required.

```
// Request Body
{
  "name": "Sunday Nwosu",
  "email": "sunday.nwosu@bemsfarms.com",
  "phone": "08078901234",
  "vehicle_type": "motorcycle",
  "zone_id": 1,
  "commission_per_delivery": 500
}

// Success Response (201 Created)
{
  "message": "Driver invitation sent successfully to sunday.nwosu@bemsfarms.com",
  "inviteUrl": "https://bemsfarms.com/driver/onboarding?token=a83f98c2...",
  "temporaryPin": "893412"
}
```

4.2 PATCH /api/admin/deliveries/drivers/:id/compliance

Description: Approves or rejects compliance submission and triggers automated email.

Auth: Admin JWT Required.

```
// Request Body (Approve)
{ "action": "approve", "notes": "All identity & vehicle docs verified clean." }

// Request Body (Reject / Request Fix)
{ "action": "reject", "notes": "Please re-upload clearer copy of driver's license." }

// Success Response (200 OK)
{
  "message": "Driver compliance approved and account activated successfully.",
  "driver": { "id": 12, "status": "active", "onboarding_status": "approved" }
}
```

4.3 GET /api/admin/deliveries/drivers

Query Params: compliance_status (invited | documents_submitted | approved | rejected), status, zone_id.

Returns: List of drivers with complete candidate dossiers, document scan URLs, and compliance summary counts.

5. Driver Self-Service Onboarding APIs

5.1 GET /api/driver/onboarding/verify?token=...

Description: Validates candidate invitation token and returns prefilled registration data.

Auth: Public (Token query parameter).

5.2 POST /api/driver/upload

Description: Multipart upload for document scans (JPEG/PNG/PDF up to 10MB). Returns hosted URL.

5.3 POST /api/driver/onboarding/submit

Description: Submits driver details, Next of Kin, Guarantor, Bank Account, and Document URLs for compliance review.

```
// Request Body
{
  "token": "a83f98c2d149...",
  "name": "Sunday Nwosu",
  "phone": "08078901234",
  "nin_number": "12984920194",
  "address": "14 Market Road, Aba",
  "emergency_contact_name": "Ngozi Nwosu",
  "emergency_contact_phone": "08033221100",
  "emergency_contact_relationship": "Spouse",
  "guarantor_name": "Chief Emeka Okafor",
  "guarantor_phone": "08099887766",
  "guarantor_address": "8 Factory Road, Aba",
  "vehicle_type": "motorcycle",
  "vehicle_plate": "ABA-492-XA",
  "license_number": "DL-90821-NG",
  "bank_name": "Access Bank",
  "account_number": "0123456789",
  "account_name": "Sunday Nwosu",
  "documents": {
    "drivers_license": "https://api.bemsfarms.com/uploads/drivers/lic_12.jpg",
    "nin_slip": "https://api.bemsfarms.com/uploads/drivers/nin_12.jpg",
    "vehicle_registration": "https://api.bemsfarms.com/uploads/drivers/reg_12.jpg",
    "passport_photo": "https://api.bemsfarms.com/uploads/drivers/photo_12.jpg",
    "guarantor_form": "https://api.bemsfarms.com/uploads/drivers/guar_12.jpg"
  }
}
```

6. Driver Mobile App Operational APIs

Method & Endpoint	Auth	Description
POST /api/driver/auth/login	None	Login with Phone & PIN. Rejects unapproved drivers with 403.
GET /api/driver/auth/me	Bearer	Returns driver profile, wallet balance, and rating.
PATCH /api/driver/auth/availability	Bearer	Toggles shift Online / Offline (is_available: true/false).
POST /api/driver/location	Bearer	Streams GPS coordinates (lat, lng, heading, speed) every 10–15s.
GET /api/driver/deliveries/available	Bearer	Feed of pending orders ready for dispatch in driver's zone.
GET /api/driver/deliveries/active	Bearer	Current order in progress with navigation waypoints.
POST /api/driver/deliveries/:id/accept	Bearer	Driver accepts delivery assignment.
PATCH /api/driver/deliveries/:id/status	Bearer	Milestone updates: assigned → awaiting_pickup → en_route → arrived → delivered.
GET /api/driver/wallet/history	Bearer	Driver delivery commissions, bonuses, and withdrawal log.
POST /api/driver/wallet/withdraw	Bearer	Requests commission payout to saved bank account.

7. Automated Email Notification Matrix

Template Function	Trigger Event	Payload & Content
sendDriverInvitationEmail	Admin sends onboarding invite	7-day onboarding portal link, temporary PIN, vehicle type, and BemsFarms welcome guide.
sendDriverApprovedEmail	Admin approves compliance	Compliance approval announcement, driver Phone/PIN credentials, and direct app launch button.
sendDriverRejectionEmail	Admin requests correction	Compliance feedback notes detailing missing/blurry items with 1-click re-upload button.

Document Authority: Bems Farms Logistics & Platform Architecture

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